MASTER CLASS DELIVERABLE • VIDEO #15

2D to 3D AI Masterclass: Generate Game-Ready 3D Assets

Microsoft Trellis, TripoSR & 1-Click Blender Automation Pipeline

AI ENGINE: MICROSOFT TRELLIS / TRIPOSR 3D SOFTWARE: BLENDER 4.2+ / PYTHON BPY PUBLISHED BY: @PARKER.RENDER

1. The 2026 3D Generative AI Revolution

Manual 3D modeling from scratch—box modeling, retopology, UV unwrapping, baking ambient occlusion and normal maps—takes days per asset. With **Microsoft Trellis** and **TripoSR**, a single front-facing photograph or AI concept art is transformed into a textured, high-resolution GLB/OBJ mesh in under 10 seconds.

2. 1-Click Blender Auto-Import Script (import_ai_mesh.py)

```
import bpy
import os

def import_and_optimize_ai_mesh(filepath):
    # Clear default cube
    if "Cube" in bpy.data.objects:
        bpy.data.objects.remove(bpy.data.objects["Cube"], do_unlink=True)

    # Import GLB/OBJ
    if filepath.endswith('.glb') or filepath.endswith('.gltf'):
        bpy.ops.import_scene.gltf(filepath=filepath)
    elif filepath.endswith('.obj'):
        bpy.ops.wm.obj_import(filepath=filepath)

    obj = bpy.context.selected_objects[0]
    bpy.context.view_layer.objects.active = obj

    # Auto-smooth normals & Add Decimate modifier for games
    bpy.ops.object.shade_smooth()
    mod = obj.modifiers.new(name="Decimate_Optimization", type='DECIMATE')
    mod.ratio = 0.5  # Reduce 50% polygon count for realtime 60FPS engines

    print(f"✅ AI Mesh successfully imported and optimized: {obj.name}")

# Run directly inside Blender Scripting Workspace
import_and_optimize_ai_mesh("D:/assets/trellis_generated_prop.glb")
```

3. Key Production Best Practices

- **Input Image Quality:** Use clean white or transparent backgrounds (remove.bg) for sharp silhouettes.
- **PBR Texture Extraction:** Trellis generates albedo, normal, and roughness maps automatically.
- **Unreal Engine / Unity Ready:** Export as compressed glTF / USDZ for immediate game engine placement.

Crafted by **@parker.render** • Visit parker-render.netlify.app/3d for updates.